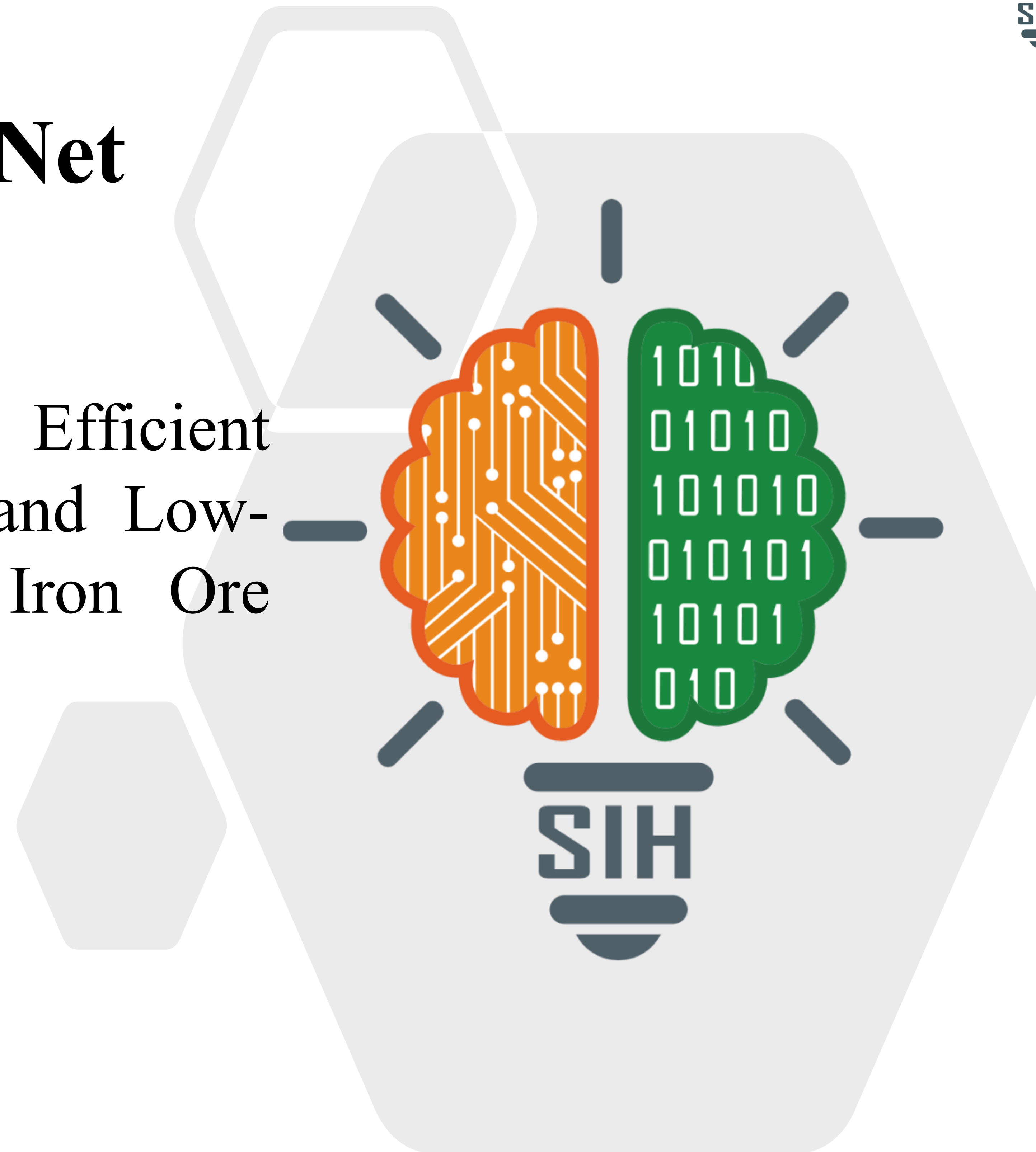


FogNet

- **Problem Statement ID** – 26007
- **Problem Statement Title**- Safe and Efficient Operation of Mine Vehicles in Fog and Low-Visibility Conditions in Open Cast Iron Ore Mines.
- **Theme**- Smart Automation
- **PS Category**- Hardware
- **Team ID**-
- **Team Name** -NEXORA



What Is The Problem ?



“Develop a real-time AI-based safety system for mine vehicles to detect fog, obstacles, and hazards, and provide advance warnings to prevent accidents in low-visibility conditions.”

❖ Proposed Solution

Radar-First Detection: 77 GHz radar detects and tracks vehicles and obstacles even in dense fog.

Dynamic Safe Speed: Calculates safe speed using stopping distance, braking, road grade, payload, and route conditions.

GNSS/IMU & V2V: Tracks vehicle position and shares movement data for early conflict warnings.

Road Visibility Monitoring: Roadside nodes continuously monitor visibility and weather conditions.

Driver HMI & Alerts: Provides safe-speed, hazard direction, and real-time audio, visual, and vibration alerts.

Fail-Safe Operation: Maintains local radar-based safety with a conservative degraded mode during sensor or network failures.

Innovate & Uniqueness



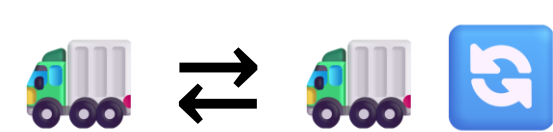
**Radar Detection
Alert**



AI sensor Fusion



V2V awareness



Alpha call



Predictive Fog



FogNet – Technology Used



1. SENSING & PERCEPTION LAYER

- Radar (primary ranging)
- Thermal camera
- NIR + RGB cameras
- GPS + IMU (position & motion)
- V2V communication via LoRa



2. INTELLIGENCE & SAFETY LAYER

- AI-based sensor fusion
- Dynamic safe-speed calculation
- Hazard & bench-edge detection
- Health monitoring & watchdog



3. COMMUNICATION LAYER

- LoRa for fog network & V2V
- Offline-first operation
- Local radio mesh between vehicles and stations



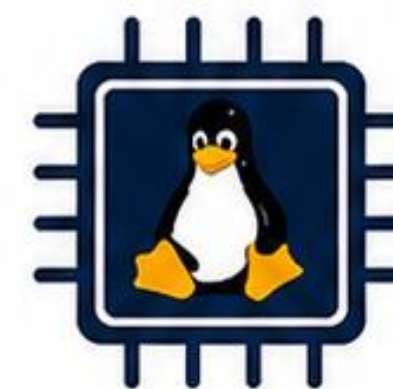
4. CONTROL & MONITORING LAYER

- Fleet dashboard for supervision
- Real-time alerts & event logging
- Emergency broadcast
- Alpha Mode escort management

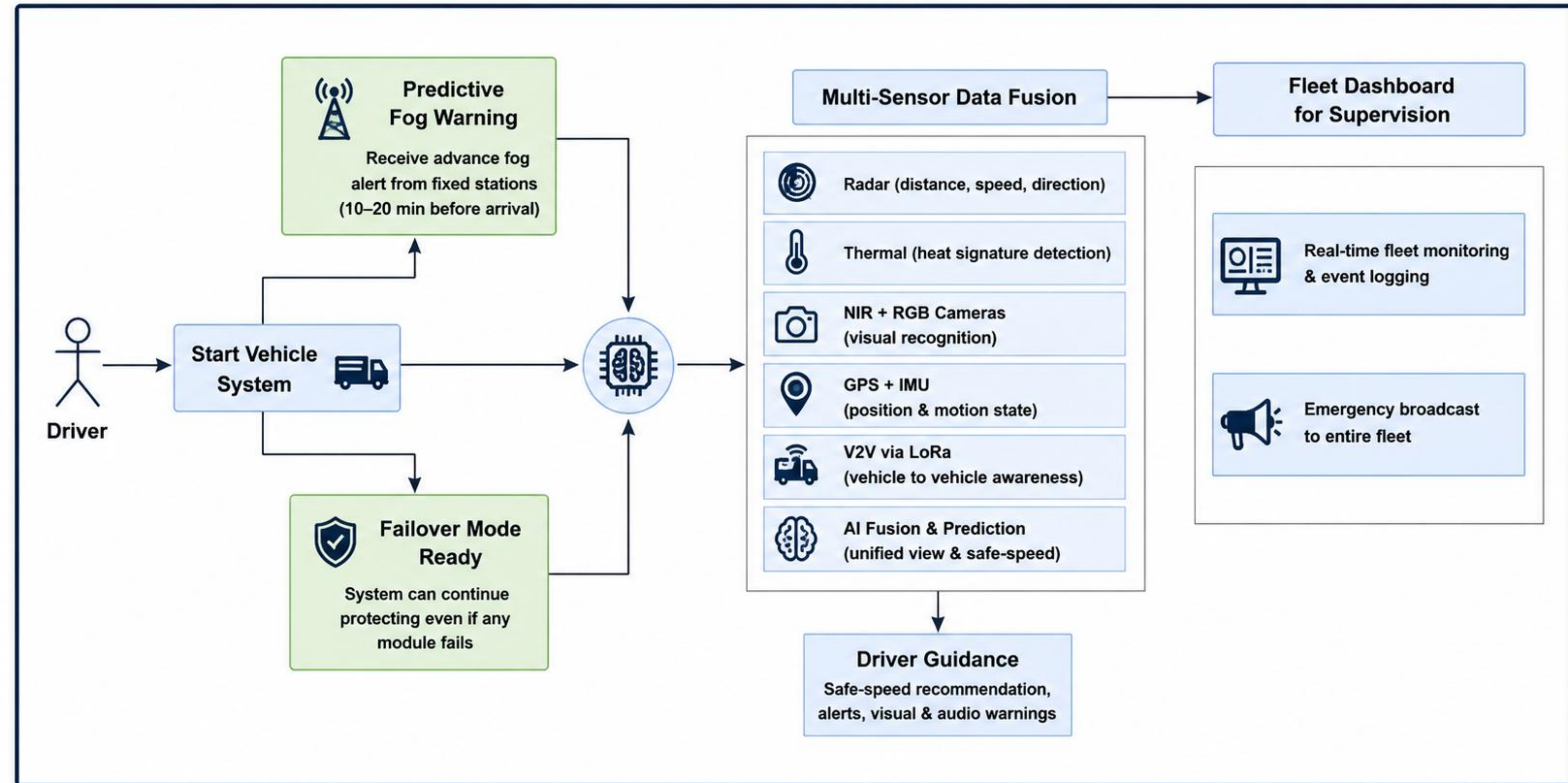


5. COMPUTING & SOFTWARE LAYER

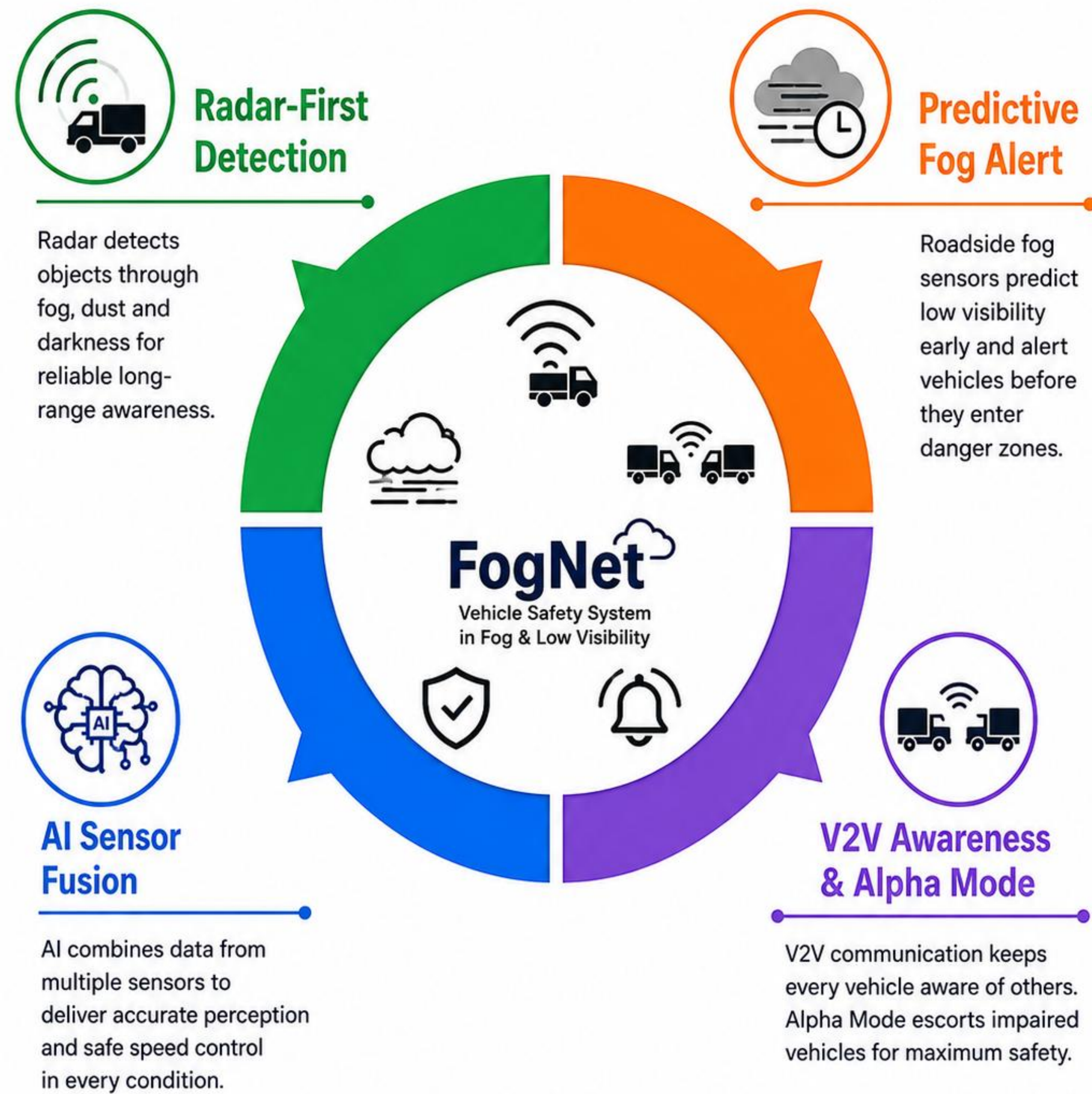
- Onboard edge processing
- Embedded Linux platform
- Optimized for low-latency safety decisions
- Human-in-the-loop interface



FogNet Work Flow



Feasibility Of the Idea



Improved Safety
Reduces accidents and saves lives.



Lower Downtime
Less disruption, more productivity.



Better Efficiency
Optimized speed and smoother operations.



Worker Protection
Detects people and hazards even in low visibility.



POTENTIAL CHALLENGES AND STRATEGIES FOR OVERCOMING



DENSE FOG & DUST

Severe fog and dust reduce visibility and detection.



RADAR + THERMAL DETECTION

Radar-first detection with thermal sensing ensures reliable detection in all conditions.



SENSOR RELIABILITY

Individual sensors may fail or give inaccurate data.



AI SENSOR FUSION

AI-based sensor fusion continuously evaluates and relies on the most reliable sensors.



UNKNOWN HAZARDS

New or unmapped hazards may not exist in the pre-loaded hazard map.



MULTI-SENSOR DETECTION

Real-time radar and multi-sensor detection identifies unknown hazards on the road.



VEHICLE SENSOR FAILURE

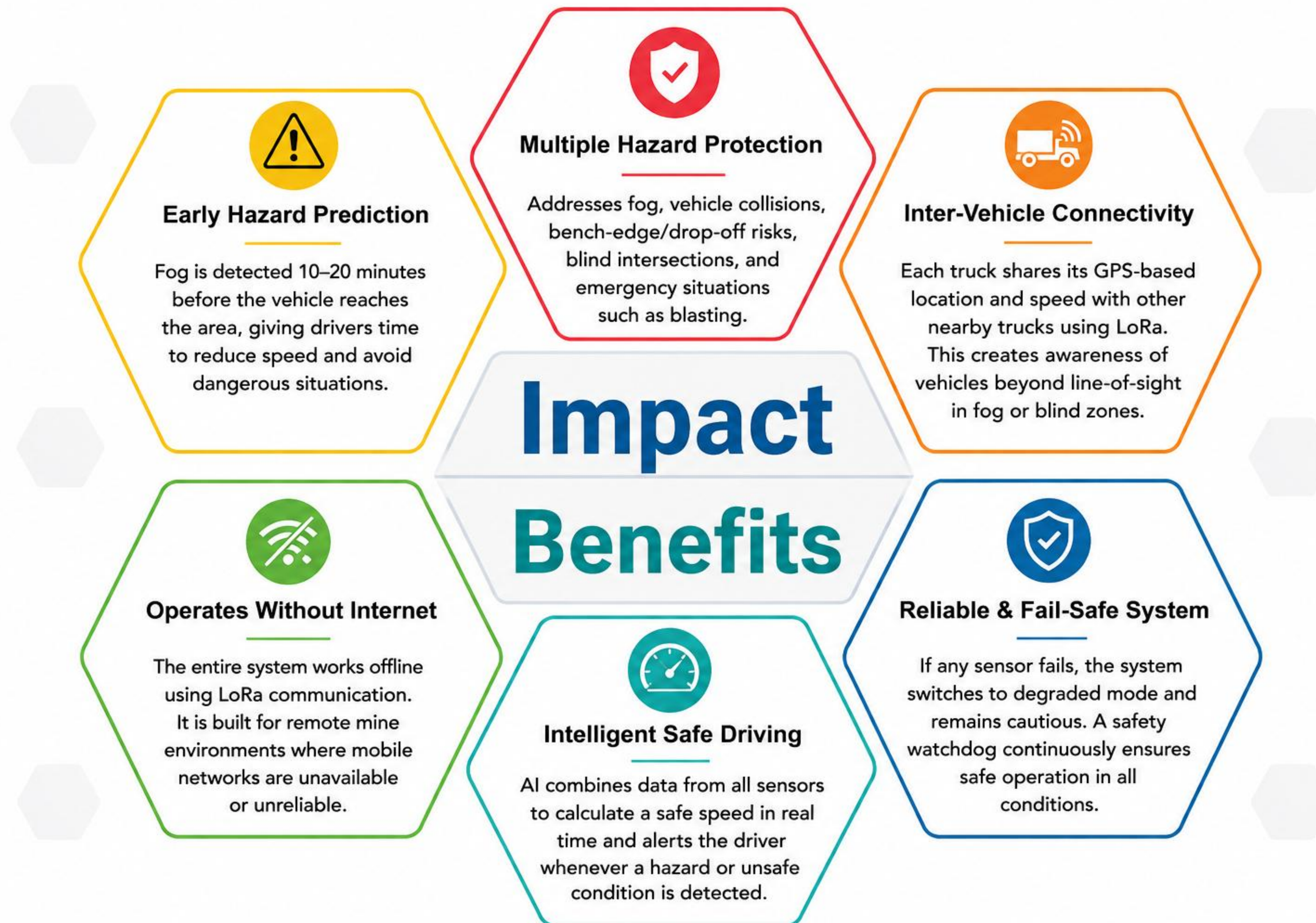
A vehicle may lose its own sensing capability.



DEGRADED SAFETY MODE

Degraded mode reduces speed, uses V2V awareness and alerts supervisor for assistance.

IMPACT AND BENEFITS



References



Feature / Aspect	FogNet (Our Solution)	Radar-Based System	LiDAR-Based System
Dense Fog Operation	✓	✓	✗
Predictive Fog Warning	✓	✗	✗
AI Sensor Fusion	✓	✗	✗
Dynamic Safe Speed Calculation	✓	✗	✗
V2V Awareness	✓	✗	✗
Alpha Escort Mode	✓	✗	✗